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A+ UNDERSTANDING AND MAINTAINANCE

DAY-1

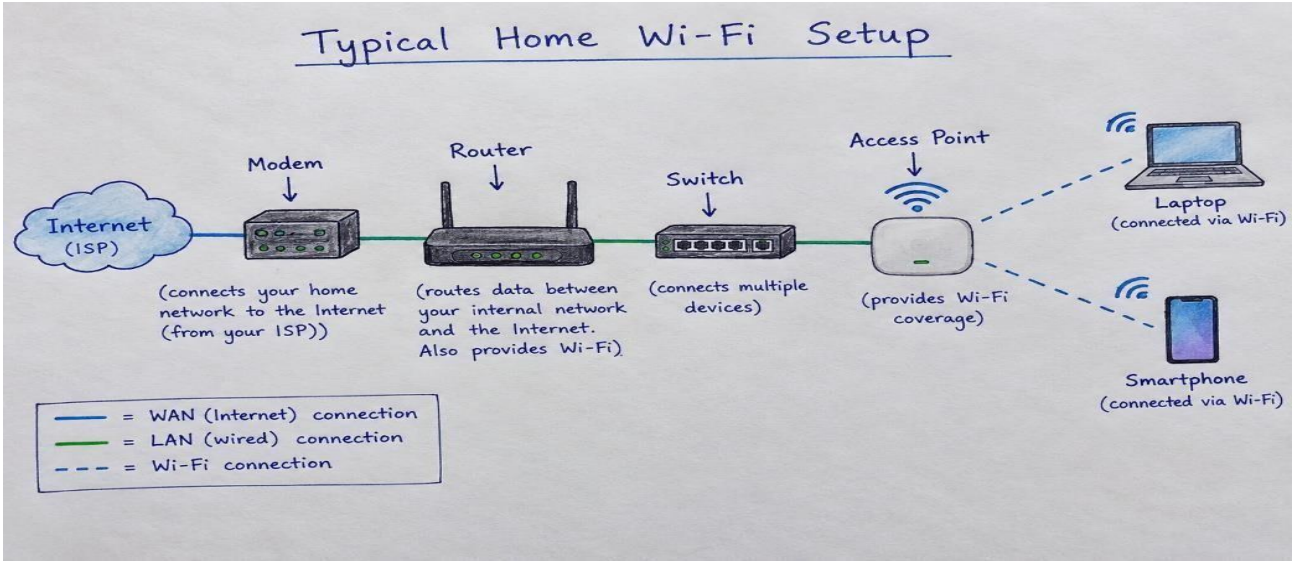
➡ TOPIC

- **INTRODUCTION TO NETWORK HARDWARE THEORY**
- **OVERVIEW OF MODEMS,ROUTERS,SWITCHES, ACCESS POINT,AND NETWORK CABLES.**

TASKS:-1

Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

RESULT



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TASKS:-2

List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

RESULT

DEVICE	MAIN FUNCTION	REAL LIFE EXAMPLE
MODEM	Connects your network to the Internet through your ISP.	Used at home to receive Internet service from Jio, Airtel, etc.
ROUTER	Routes data between your local network and the Internet and often provides Wi-Fi.	Used at home to share Internet with phones, laptops, and TV
SWITCH	Connects multiple wired devices on the same LAN	Used in an office to connect computers and printers using Ethernet.
ACCESS PT (AP)	Provides Wi-Fi access and connects wireless devices to a wired network	Used in an office, school, or large home to extend Wi-Fi coverage.
NETWORK CABLE	Carries data between network devices using a wired connection.	Ethernet cable used to connect a computer to a router or switch.

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TASKS:-3

Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

RESULT

Twisted Pair (Ethernet cable)

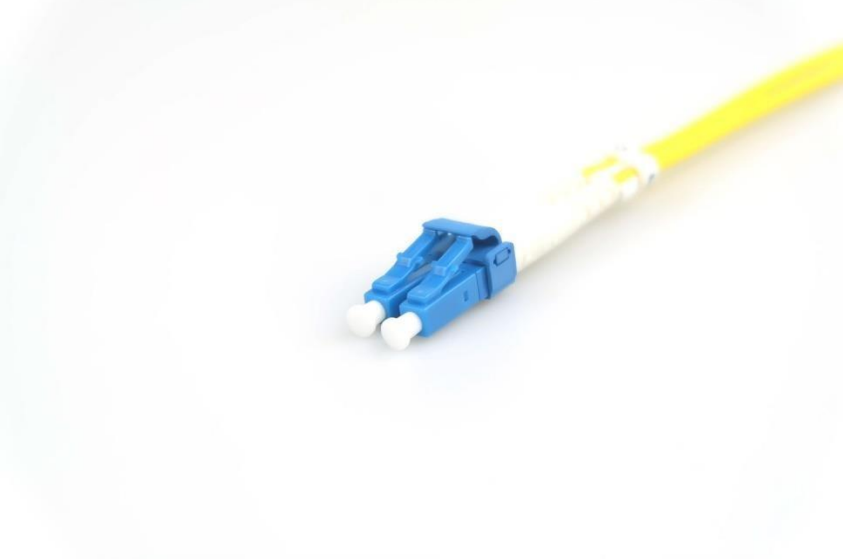


Coaxial Cable



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Fiber Optic Cable



NO.	Network cable	Comman use
1.	Ethernet Cable (RJ-45)	Commonly used to connect computers, routers, switches, and other devices in LAN networks.
2.	Coaxial Cable	Commonly used for cable TV, CCTV systems, and some broadband internet connections.
3.	Fiber Optic Cable	Commonly used for high-speed internet and long-distance communication because it carries data using light.

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Tasks:-4

Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

RESULT

Basis	**Router**	**Switch**
Main Use	Connects different networks and provides Internet access.	Connects multiple devices within the same network (LAN).
Typical Location	Usually placed between the local network and the Internet/modem.	Usually placed inside a LAN, such as an office, school, or computer lab
How It Handles Data	Forwards data packets between different networks using IP addresses.	Forwards data frames to the correct device using MAC addresses.
Network Layer	Mainly works at **Layer 3 (Network Layer)**.	Mainly works at **Layer 2 (Data Link Layer)**
Example	Home Wi-Fi router connects phones and computers to the Internet.	A switch connects several computers in a college computer lab.